

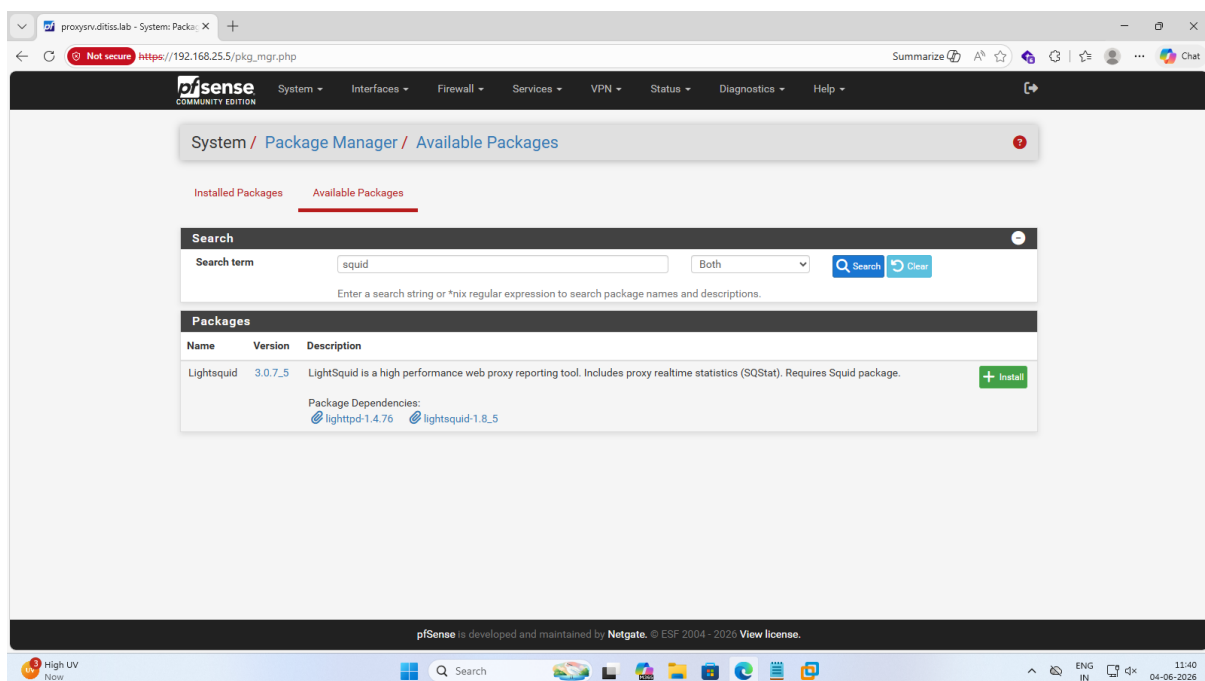
Q1.

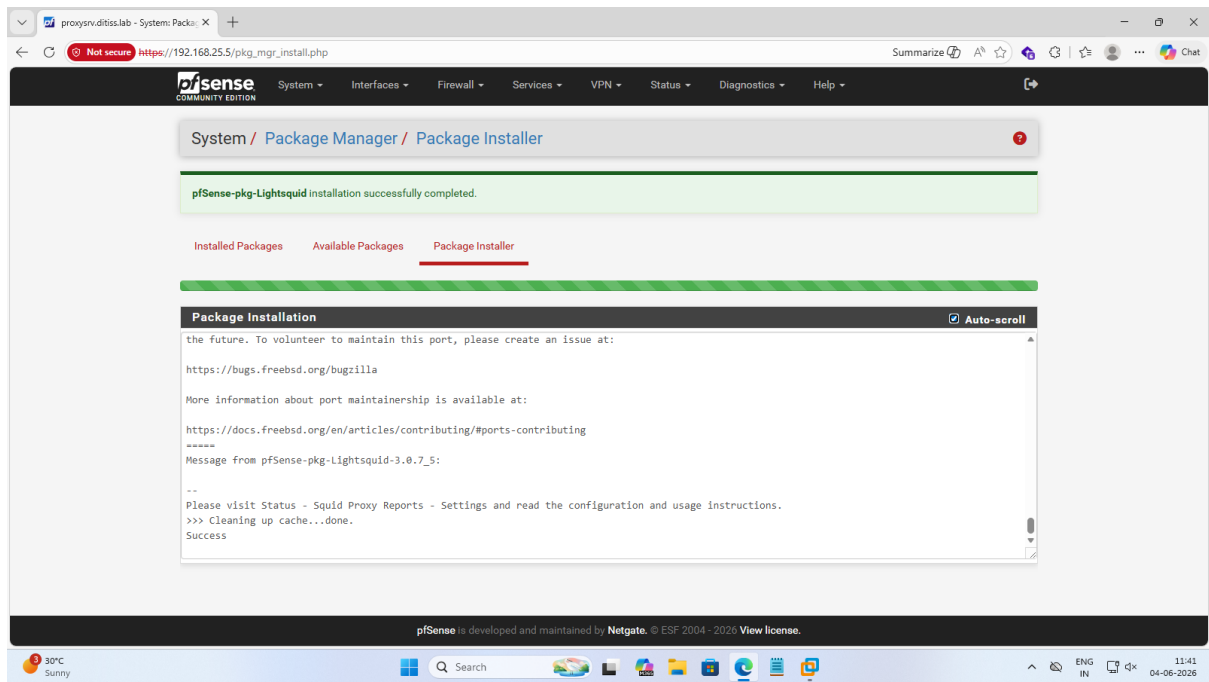
install and configure pfSense firewall and squid and squidguard. create a user by name ditiss allow centos, redhat, amazon, Microsoft, cloud website, kali linux website. create another user by name dbda. allow python, oracle, ibm, microsoft, websites. for all users allow google, cdac, bing, gmail websites, and to all users allow social networking sites for 3 hour only.

ANS ->

Step1. First do reset to factory default then check ips are in out network then open browser and put that ip in URL and do login

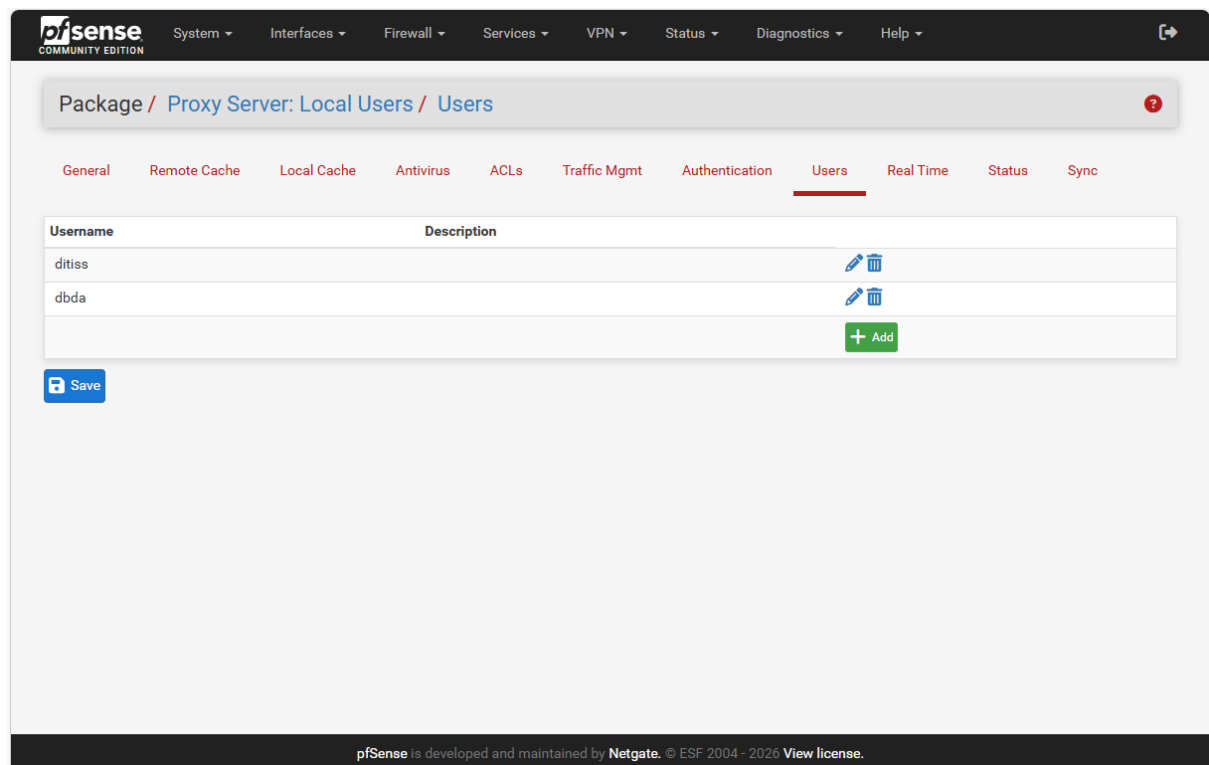
Step2. Install squid, squid guard Packages from system->Package Manager





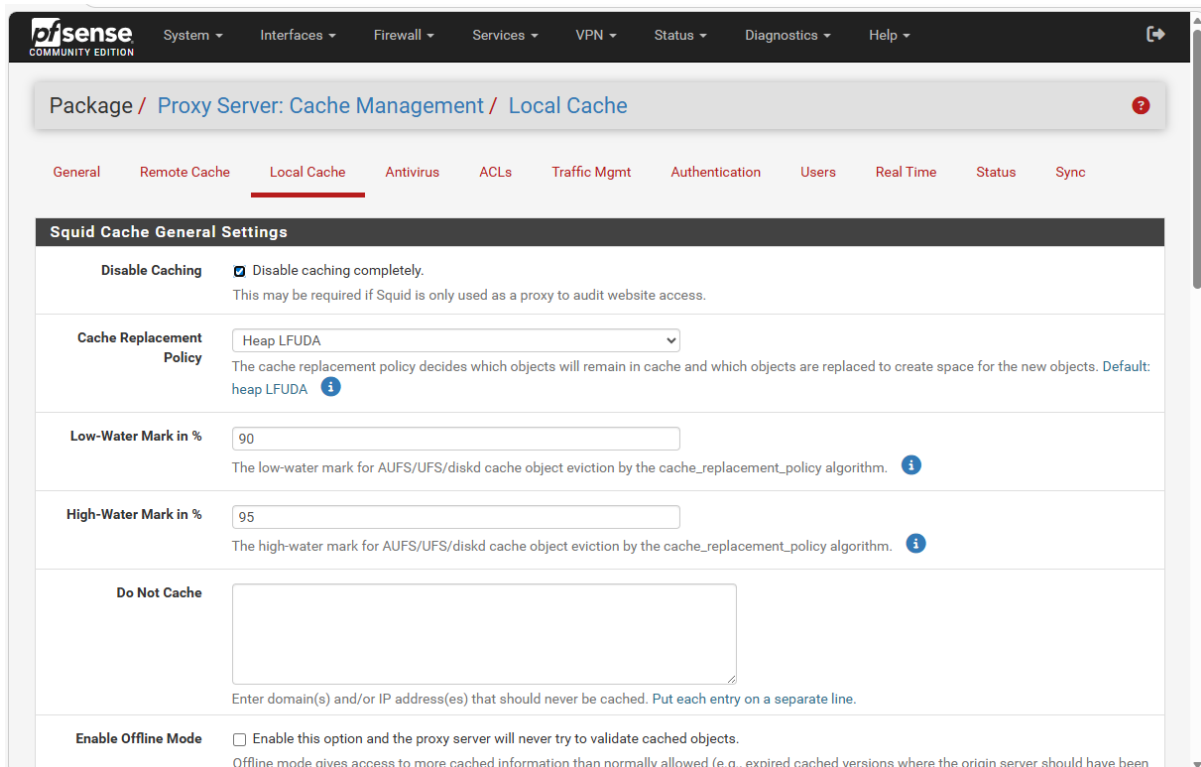
Step3. Add users ditiss & dbda

Services->squid proxy server->users



->Local Cache

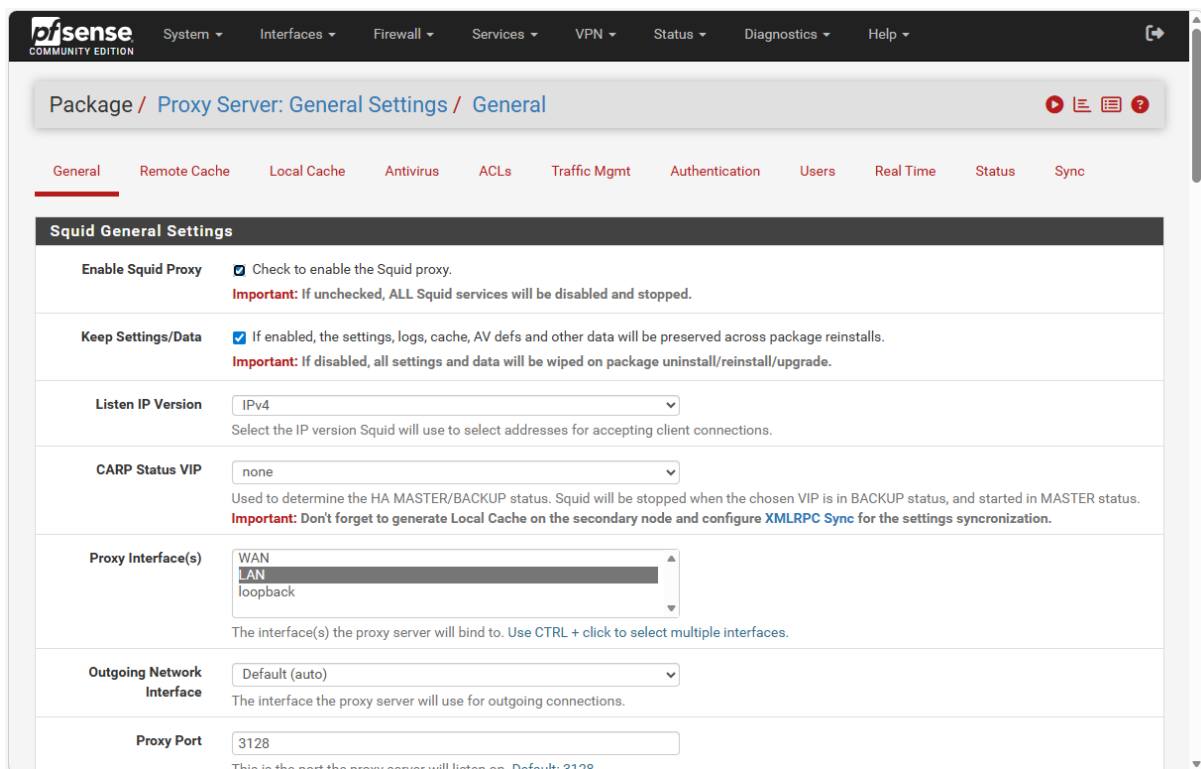
[/]Disable caching completely->save



[/]check to enable the squid proxy

[/]if enabled, the settings, logs, cache, AV defs and other data will be...

Proxy Interface-> LAN



Step4. ->services->squid guard proxy filter->Target categories

Add user and domains which are allowed

Pfsense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settings Common ACL Groups ACL **Target categories** Times Rewrites Blacklist Log XMLRPC Sync

General Options

Name
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List
Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

URL List

Add second use and domains allowed

Pfsense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settings Common ACL Groups ACL **Target categories** Times Rewrites Blacklist Log XMLRPC Sync

General Options

Name
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List
Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

URL List

Add third for common allowed name and domain

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settings Common ACL Groups ACL **Target categories** Times Rewrites Blacklist Log XMLRPC Sync

General Options

Name whitelist
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order dbda
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List google.com cdac.in bing.com
Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

URL List

Step5.

->Groups ACL

Add group name and add client name in 'ditiss'

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Proxy filter SquidGuard: Groups Access Control List (ACL) / Edit / Groups ACL

General settings Common ACL **Groups ACL** Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Disabled ☐ Check this to disable this ACL rule.

Name ditiss_sites
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order -----
Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note: Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example: ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source) 'ditiss'
Enter client's IP address or domain or "username" here. To separate them use space.
Example:
IP: 192.168.0.1 - Subnet: 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - IP-Range: 192.168.1.1-192.168.1.10
Domain: foo.bar matches foo.bar or *.foo.bar
Username: 'user1'
Ldap search (Ldap filter must be enabled in General Settings):
ldapusersearch ldap://192.168.0.100/DC=domain,DC=com?sAMAccountName=sub?(&(sAMAccountName=%s)(memberOf=CN=#2aCN=User%2aDC=domain%2aDC=com))

Add target Rules list allowed only user ditiss and whitelist

Whitelist is for common so add it and Default access Deny in Target Category only don't do anything at other side.

[/]check for logs.

Target Rules

whitelist ditiss !all [all]

Target Rules List + -

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories

Target Categories for off-time

If 'Time' not defined, this is column will be ignored.

[dbda]	access	---	[dbda]	access	---
[social_sites]	access	---	[social_sites]	access	---
[whitelist]	access	allow	[whitelist]	access	---
[ditiss]	access	allow	[ditiss]	access	---
Default access [all]	access	deny	Default access [all]	access	allow

Do not allow IP-Addresses in URL

☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Redirect mode

none

Select redirect mode here.
Note: if you use 'transparent proxy', then 'int' redirect mode will not accessible.
Options: [ext url err page](#), [ext url redirect](#), [ext url as 'move'](#), [ext url as 'found'](#).

Redirect

Enter the external redirection URL, error message or size (bytes) here.

Use SafeSearch engine

☐ To protect your children from adult content you can use the protected mode of search engines.
At the moment it is supported by Google, Yandex, Yahoo, MSN, Live Search, Bing, DuckDuckGo, OneSearch, Rambler, Ecosia and Qwant. Make sure that the search engines can be accessed. It is recommended to prohibit access to others.
Note: This option overrides 'Rewrite' setting.

Rewrite

none (rewrite not defined)

Enter the rewrite condition name for this rule or leave it blank.

Rewrite for off-time

none (rewrite not defined)

Do same thing for second user


System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Proxy filter SquidGuard: Groups Access Control List (ACL) / Edit / Groups ACL

General settings Common ACL **Groups ACL** Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Disabled ☐ Check this to disable this ACL rule.

Name
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order
Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note:
Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example:
ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source)
Enter client's IP address or domain or "username" here. To separate them use space.
Example:
IP: 192.168.0.1 - **Subnet:** 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - **IP-Range:** 192.168.1.1-192.168.1.10
Domain: foo.bar matches foo.bar or *.foo.bar
Username: 'user1'
Ldap search (Ldap filter must be enabled in General Settings):
ldapusersearch ldap://192.168.0.100/DC=domain,DC=com?sAMAccountName?sub?(&(sAMAccountName=%s)(memberOf=CN=it%2cCN=Users%2cDC=domain%2cDC=com))
Attention: these line don't have break line, all on one line

Time

Target Rules

Target Rules List + -

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories	Target Categories for off-time
If 'Time' not defined, this column will be ignored.	
[dbda]	[dbda]
[social_sites]	[social_sites]
[whitelist]	[whitelist]
[ditiss]	[ditiss]
Default access [all]	Default access [all]

Do not allow IP-Addresses in URL ☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Redirect mode
Select redirect mode here.
Note: if you use 'transparent proxy', then 'int' redirect mode will not accessible.
Options: [ext url err page](#), [ext url redirect](#), [ext url as 'move'](#), [ext url as 'found'](#).

Redirect
Enter the external redirection URL, error message or size (bytes) here.

Use SafeSearch engine ☐ To protect your children from adult content you can use the protected mode of search engines.
At the moment it is supported by Google, Yandex, Yahoo, MSN, Live Search, Bing, DuckDuckGo, OneSearch, Rambler, Ecosia and Qwant. Make sure that the search engines can be accessed. It is recommended to prohibit access to others.
Note: This option overrides 'Rewrite' setting.

Rewrite
Enter the rewrite condition name for this rule or leave it blank.

Rewrite for off-time

->Common ACL

Allowed to only whitelist [/]donot allow ip-addresses in URL [/] logs

The screenshot shows the pfSense web interface for configuring the Proxy filter SquidGuard: Common Access Control List (ACL). The breadcrumb trail is Package / Proxy filter SquidGuard: Common Access Control List (ACL) / Common ACL. The navigation bar includes links for General settings, Common ACL (selected), Groups ACL, Target categories, Times, Rewrites, Blacklist, Log, and XMLRPC Sync.

General Options

Target Rules
whitelist !all

Target Rules List (+) (-)
ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories

Category	Access
[whitelist]	allow
[dntda]	---
[ditiss]	---
Default access [all]	deny

Do not allow IP-Addresses in URL
☒ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Proxy Denied Error

The first part of the error message displayed to clients when access was denied. Defaults to **Request denied by g_get('product_name') proxy**.

Redirect mode
int error page (enter error message)
Select redirect mode here.
Note: if you use 'transparent proxy', then 'int' redirect mode will not be accessible.
Options: [ext url err page](#), [ext url redirect](#), [ext url as 'move'](#), [ext url as 'found'](#).

Redirect info

Step.6

->Target categories

Add social media sites for 3 hours [/]logs

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settings Common ACL Groups ACL **Target categories** Times Rewrites Blacklist Log XMLRPC Sync

General Options

Name
 Enter a unique name of this rule here.
 The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order
 Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List
 Enter destination domains or IP-addresses here. To separate them use space.
 Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

URL List

->Time

Check current time on clicking logo of pfsense from that time calculate for 3 hours here provide only name and add only time range->save

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Proxy filter SquidGuard: Times / Edit / Times

General settings Common ACL Groups ACL Target categories **Times** Rewrites Blacklist Log XMLRPC Sync

General Options

Name
 Enter a unique name of this rule here.
 The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Values
 Time type Days Date or Date range Time range

Add

Description
 You may enter any description here for your reference.
Note:
 Example for Date or Date Range: 2007.12.31 or 2007.11.31-2007.12.31 or *.12.31 or 2007.*.31
 Example for Time Range: 08:00-18:00

->Groups ACL->ad name social_sites add users 'ditiss' 'dbda' add socialmedia in time

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Proxy filter SquidGuard: Groups Access Control List (ACL) / Edit / Groups ACL ?

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Disabled ☐ Check this to disable this ACL rule.

Name
 Enter a unique name of this rule here.
 The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order
 Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note:
 Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example:
 ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source)
 Enter client's IP address or domain or "username" here. To separate them use space.
Example:
 IP: 192.168.0.1 - Subnet: 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - IP-Range: 192.168.1.1-192.168.1.10
 Domain: foo.bar matches foo.bar or *.foo.bar
 Username: 'user1'
 Ldap search (Ldap filter must be enabled in General Settings):
 ldapusersearch ldap://192.168.0.100/DC=domain,DC=com?sAMAccountName=sub?(&(sAMAccountName=%s)
 (memberOf=CN=it%2cCN=Users%2cDC=domain%2cDC=com))
 Attention: these line don't have break line, all on one line

Time

Allow social_sites and whitelist in Target Rule list [/]donot allowed ip [/]logs->save

Time social_sites

Select the time in which 'Target Rules' will operate or leave 'none' for rules without time restriction. If this option is set then in off-time the second ruleset will operate.

Target Rules social_sites whitelist !all [social_sites whitelist !all]

Target Rules List

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories			Target Categories for off-time If 'Time' not defined, this is column will be ignored.		
[social_sites]	access	allow	[social_sites]	access	allow
[whitelist]	access	allow	[whitelist]	access	allow
[dbda]	access	—	[dbda]	access	—
[dntss]	access	—	[dntss]	access	—
Default access [all]	access	deny	Default access [all]	access	deny

Do not allow IP-Addresses in URL ☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Redirect mode none

Select redirect mode here.
 Note: if you use 'transparent proxy', then 'int' redirect mode will not accessible.
 Options: [ext url err page](#), [ext url redirect](#), [ext url as 'move'](#), [ext url as 'found'](#).

Redirect

Enter the external redirection URL, error message or size (bytes) here.

Use SafeSearch engine ☐ To protect your children from adult content you can use the protected mode of search engines.
 At the moment it is supported by Google, Yandex, Yahoo, MSN, Live Search, Bing, DuckDuckGo, OneSearch, Rambler, Ecosia and Qwant. Make sure that the search engines can be accessed. It is recommended to prohibit access to others.
Note: This option overrides 'Rewrite' setting.

Rewrite none (rewrite not defined)

Enter the rewrite condition name for this rule or leave it blank.

Rewrite for off-time none (rewrite not defined)

Enter the rewrite condition name for this rule or leave it blank.

Description

You may enter any description here for your reference.

Log ☒ Check this option to enable logging for this ACL.

Save

Step.7

->General settings

Apply->save

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Package / Proxy filter SquidGuard: General settings / General settings

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Enable ☒ Check this option to enable squidGuard.
Important: Please set up at least one category on the 'Target Categories' tab before enabling. See [this link for details](#).
 The Save button at the bottom of this page must be clicked to save configuration changes.
 To activate squidGuard configuration changes, **the Apply button must be clicked.**

SquidGuard service state: **STARTED**

LDAP Options

Enable LDAP Filter ☐ Enable options for setup ldap connection to create filters with ldap search

LDAP DN
 Configure your LDAP DN (ex: cn=Administrator,cn=Users,dc=domain)

LDAP DN Password
 Password must be initialize with letters (Ex: Change123), valid format: [a-zA-ZV][a-zA-Z0-9/_\.\V\%\+\=?=&]

LDAP Cache Time
 Number of seconds to cache LDAP Results (recommended value: 300)

Strip NT domain name ☐ Strip NT domain name component from user names (/ or \ separated).

Step.8

->Services->Squid Proxy server->Authentication

Authentication method->Local->save

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Package / Proxy Server: Authentication / Authentication

General Remote Cache Local Cache Antivirus ACLs Traffic Mgmt **Authentication** Users Real Time Status Sync

Squid Authentication General Settings

Authentication Method
 Select an authentication method. This will allow users to be authenticated by local or external services.

Authentication Server
 Enter the IP or hostname of the server that will perform the authentication here.

Authentication server port
 Enter the port to use to connect to the authentication server here.
 Leave this field blank to use the authentication method's default port.

Authentication Prompt
 This string will be displayed at the top of the authentication request window.

Authentication Processes
 The number of authenticator processes to spawn. If many authentications are expected within a short timeframe, increase this number accordingly.

Authentication TTL
 This specifies for how long (in minutes) the proxy server assumes an externally validated username and password combination to be valid.
 When the Time To Live expires, the user will be prompted for credentials again. **Default: 5**

Authentication Max User IP Addresses
 Enforces a limit to the number of unique IP addresses from which a single user can login.
 Attempts to login from additional IP addresses are denied until the Authentication TTL has expired

Check on each user all configuration is done well.